

Fabio Arnez

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Experience

Research Engineer in AI — Trustworthy/Safe Deep Learning Team Leader *Palaiseau, France*
Commissariat à l'énergie atomique et aux énergies alternatives (CEA), DRT.LIST.DILS.LSEA *Oct 2019 – present*
 Research on DNNs/LLMs/VLMs/VLAs confidence representation (uncertainty estimation) for out-of-distribution/hallucination detection, runtime monitoring in complex computer vision tasks ([demo object detection](#)), quantization and embedded uncertainty estimation, safe and robust end-to-end DNN-based [UAV navigation](#), and AI Safety.

- Ph.D. theses and internships supervision
- Developed [RunIA-Core](#), a Python library for DNN confidence representation and runtime monitoring, with support for PyTorch and PyTorch-Lightning, and integration with MLFlow for experiment tracking
- Contributed to the PRIMaL project: Developed an uncertainty-based method for LLM extrinsic & intrinsic hallucination detection, developed OoD object detection methods, collected a dataset for UAV navigation using the PyBullet simulator (gym-pybullet-drones)
- Contributing to the [DeepGreen](#) project for embedded AI, (France 2030 program): Developed dropout operator for the [Aidge](#) library for embedded DNN uncertainty estimation, studied the reliability & robustness of quantized CLIP
- Contributed to the [Confiance.ai](#) program, (France 2030 program): Developed an OoD detection method [library](#)
- Contributed to the [Comp4Drones](#) project (H2020 Program European Union): Led WP4-Task4.4 on UAV Runtime Monitoring, developed a distributed uncertainty runtime monitoring system for DNN-based UAV navigation systems

Data Scientist *Cochabamba, Bolivia*
Bamboo *Apr 2019 – Oct 2019*

- Engineering team leader; enterprise data analysis & data visualization
- Implemented PowerBI dashboards to support enterprise decision-making

Research Assistant *Lugano, Switzerland*
University of Applied Arts and Science from Southern Switzerland (SUPSI), ISEA, SMT Lab *Sept 2016 – Aug 2018*
 Part-time embedded systems development

- Developed embedded machine learning prototypes for signal classification
- Developed embedded system prototypes for wireless IoT applications (LoRaWAN, 802.15.4, Bluetooth)

Unmanned Aerial Systems Researcher *Cochabamba, Bolivia*
Jalasoft *May 2014 – May 2016*
 Localization & navigation, ROS integration, embedded systems integration, computer vision tasks

- Built a custom UAV from scratch employing the PX4/Pixhawk autopilot
- Integrated software & hardware components (modular system) using ROS, and implemented onboard computer vision tasks

Adjunct Researcher and Lecturer *Cochabamba, Bolivia*
Universidad Privada Boliviana (UPB) *Sept 2014 – Mar 2019*
 Research in embedded wireless IoT applications, Lecturer

- Contributed to the Smart Street Lightning project ([SRESLi](#))
- Courses Taught (2014-2019): Embedded Electronic Systems, Microprocessor Architecture & Technology, Electronics & Telecom. Project, Telecom. Electronics, Electronic Instrumentation

Education

Université Paris-Saclay *Palaiseau, France*
Ph.D. in Computer Science - Artificial Intelligence *Nov 2019 – Dec 2023*
Thesis: Deep Neural Network Uncertainty Runtime Monitoring for Robust and Safe AI-based Automated Navigation

- DNN uncertainty-based out-of-distribution detection, uncertainty propagation for robust UAV navigation, confidence monitoring in UAV learning-based components for trustworthy navigation.

University of Applied Arts and Science from Southern Switzerland (SUPSI) *Lugano, Switzerland*
MSc. in Engineering: Embedded Systems & Microelectronics *Sept 2016 – July 2018*
 Thesis Title: *Real-Time Human Footstep Recognition on Smart Anti-Static Floor*

- Signal processing and classification using machine learning targeting resource constrained-hardware

Skills

Programming: Python (Numpy, Scipy, Matplotlib, Pandas, Seaborn...), C/C++; PyQt5 for GUI dev; Git

Machine Learning & Deep Learning: PyTorch, PyTorch-Lightning, Hugging Face, Hydra, MLFlow, Slurm HPC

Robotics: ROS/ROS 2; PyBullet, MuJoCo, CARLA, and AirSim simulators; Gymnasium; PX4/Pixhawk autopilot

Embedded Systems: ST/SiliconLabs/NXP ARM Cortex (M0+, M3, M4), FreeRTOS; Aidge

Languages: English (fluent, C1/C2), French (basic-intermediate, B1), Italian (basic), Spanish (native)

Selected Publications

- Bouguerra, A., Gomez-Villa, A., Mraidha, C., **Arnez, F.** (2026, December). *Disentangling the Good From the Bad: Quantization-Induced Flips Are Not Random*. To appear in **Conference on Neural Information Processing Systems (NeurIPS)**.
- Bouguerra, A., Radermacher, A., Mraidha, C., **Arnez, F.** (2026, November). *Quantization and Corruption Robustness in Deployed Road-Scene Perception*. To appear in IEEE International Conference on Vehicular Electronics and Safety (ICVES).
- Montoya, D., Espinoza Mayzer, M. S., **Arnez, F.** (2026, November). *Drive-SynOOD-OD: A Diffusion-Inpainted Synthetic Out-of-Distribution Object Detection Benchmark for Autonomous Driving*. To appear in IEEE International Conference on Vehicular Electronics and Safety (ICVES). [dataset](#)
- **Arnez, F.**, & Gomez-Villa, A. (2026, July). *The SIGReg Objective as Variational Free Energy: A Theoretical Active-Inference Account of JEPa World Models*. arXiv preprint arXiv:2607.13612. [link](#)
- Bouguerra, A., Montoya, D., Gomez-Villa, A., Mraidha, C., **Arnez, F.** (2026, July). *Less Precise Can Be More Reliable: A Systematic Evaluation of Quantization's Impact on VLMs Beyond Accuracy*. In **International Conference on Machine Learning (ICML)**. [link](#)
- Suri, K., & **Arnez, F.** (2026, June). *Digital Twins and World Models: A Systematic Taxonomic Disambiguation*. In International Conference on Advanced Information Systems Engineering (CAiSE). [link](#)
- Hajji, E., Bouguerra, A., **Arnez, F.** (2025, November). *The Map of Misbelief: Tracing Intrinsic and Extrinsic Hallucinations Through Attention Patterns*. In Proceedings of the AAAI Symposium Series (AAAI Fall Symposium - ATRACC), 7(1). [link](#)
- Montoya, D., Bouguerra, A., Gomez-Villa, A., & **Arnez, F.** (2025, June). *FindMeIfYouCan: Bringing Open Set Metrics to Near, Far and Farther Out-of-Distribution Object Detection*. arXiv preprint arXiv:2506.14008. (Under review) [link](#)
- **Arnez, F.**, Vasquez, D. A. M., Radermacher, A., & Terrier, F. (2024, July). *Latent Representation Entropy Density for Distribution Shift Detection*. In **Conference on Uncertainty in Artificial Intelligence (UAI)**. [link](#)
- **Arnez, F.**, Espinoza, H., Radermacher, A., & Terrier, F. (2022, September). *Towards Dependable Autonomous Systems Based on Bayesian Deep Learning Components* In 2022 18th European Dependable Computing Conference (EDCC) (pp. 65-72). IEEE. [link](#)
- **Arnez, F.**, Espinoza, H., Radermacher, A., & Terrier, F. *Improving Robustness of Deep Neural Networks for Aerial Navigation by Incorporating Input Uncertainty* Proceedings of the Workshop on Artificial Intelligence Safety Engineering WAISE 2021, International Conference on Computer Safety, Reliability, and Security (pp. 219-225). Springer. [link](#)
- **Arnez, F.**, Espinoza, H., Radermacher, A., & Terrier, F. *A Comparison of Uncertainty Estimation Approaches in Deep Learning Components for Autonomous Vehicle Applications*. Proceedings of the Workshop on Artificial Intelligence Safety 2020, vol 2640, ISSN:1673-0073. [link](#)
- Full list of publications in [Google Scholar profile](#)

Other Scientific Activities

- **Ph.D. Thesis Supervision:** *Deep Neural Network Uncertainty Estimation on Embedded Targets* (Aymen Bouguerra, 2024 - Present); *Out-of-Distribution Detection with Vision Foundation Models and Post-hoc Methods* (starting Oct. 2026)
- **Reviewer in Conferences/Workshops:** NeurIPS (2026), ICML (2026 – Gold Reviewer), BMVC (2026), CVPR (2025), AAAI (2026, 2025), UAI (2026, 2025, 2024), WACV (2025), IV (2025, 2022), AI-Safety workshop at IJCAI (2024), Releasing Robots in the Wild Workshop at ICRA (2022)
- **Workshop & Conference Organizing Committee Member:** Workshop on AI Safety and Security at IJCAI 2024, ICVES 2026 (ongoing)